

EXPERIMENT-3

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Subject Name: Computer Networks

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AIM:

To understand the concepts of:

1. Hub
2. Layer-2 Switch
3. Router
4. Difference between Hub and Switch

and simulate their working using Cisco Packet Tracer.

OBJECTIVES:

- To study the working of a Hub.
- To understand data transmission through broadcasting.
- To study the working of a Layer-2 Switch.
- To understand MAC Address Table learning.
- To study the role of a Router in connecting multiple LANs.
- To configure IP Addressing and Default Gateway.
- To verify communication between devices using Simple PDU and Ping.
- To compare Hub and Switch based on functionality and performance.

SOFTWARE REQUIREMENTS:

(a) Software:

- Cisco Packet Tracer
- Windows Operating System

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(b) Hardware (Virtual):

- PC-PT
- Hub-PT
- Cisco 2960-24TT Switch
- Cisco 2911 Router

THEORY: (i) Hub

A Hub is a networking device used to connect multiple computers in a Local Area Network (LAN). It works at the Physical Layer (Layer 1) of the OSI model and simply forwards electrical signals without examining any address information.

A hub has multiple ports and is commonly used to create a Star Topology. Whenever a packet arrives on one port, the hub copies the packet and broadcasts it to every other connected port. Every device receives the frame, but only the intended destination accepts it while the others discard it.

Since the hub broadcasts all traffic, it creates unnecessary network congestion and increases collisions. It operates only in Half-Duplex mode and does not maintain any memory or MAC address table.

Advantages

- Less expensive than switches.
- Easy to install.
- Suitable for small LANs.

Disadvantages

- Broadcasts all packets.
- No memory.
- High collision domain.
- Half-duplex communication.
- Low security.

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(ii) Layer-2 Switch

A Switch is a Layer-2 networking device that connects devices within a LAN. Unlike a hub, a switch maintains a MAC Address Table in its memory.

Whenever a frame arrives, the switch learns the source MAC address and stores it along with the port number. Later, when another frame is sent, the switch forwards it only to the destination port instead of broadcasting to all devices.

This intelligent forwarding minimizes collisions, improves bandwidth utilization, and provides better security and performance.

Features

- Layer-2 device.
- Stores MAC Address Table.
- Intelligent forwarding.
- Supports Unicast, Broadcast and Multicast.
- Full-Duplex communication.
- More efficient than Hub.

(iii) Router

A Router is a Layer-3 networking device used to connect two or more different networks. It forwards packets based on IP Addresses.

A router maintains a Routing Table that stores the best path to different networks. Each router interface belongs to a different network.

In this experiment, Router 2911 connects:

- LAN-1 → 10.0.0.0
- LAN-2 → 192.168.1.0

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The router interfaces act as the Default Gateway for their respective LANs, enabling communication between devices on different networks.

Difference between Hub, Switch and Router:

Feature	Hub	Switch	Router
OSI Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Main Function	Connect devices in LAN	Connect devices intelligently in LAN	Connect different networks
Uses	Electrical signals	MAC Addresses	IP Addresses
Address Table	No	MAC Address Table	Routing Table
Memory	No Memory	Has Memory	Has Memory
Intelligence	Not Intelligent	Intelligent	Highly Intelligent
Data Forwarding	Broadcasts to all ports	Sends only to destination MAC	Sends to destination network
Collision Domain	One	Separate for every port	Separate for every interface
Broadcast Domain	One	One (without VLAN)	Separate for each interface
Duplex Mode	Half Duplex	Full Duplex	Full Duplex
Bandwidth Sharing	Shared	Dedicated	Dedicated
Security	Low	Medium	High
Speed	Slow	Fast	Moderate (routing based)
Cost	Lowest	Medium	Highest

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Configuration	Not Required	Minimal	Complex
Can connect different LANs?	No	No	Yes
Used as Default Gateway	No	No	Yes
Routing Capability	No	No	Yes
Suitable For	Small LANs, Labs	Office/College LANs	Enterprise Networks, Internet

STEPS/IMPLEMENTATION:

Part I – Hub

1. Add one Hub.
2. Add six PCs.
3. Connect all PCs using Copper Straight-Through Cable.
4. Assign IP addresses:

PC	IP
PC0	10.10.10.1
PC1	10.10.10.2
PC2	10.10.10.3
PC3	10.10.10.4
PC4	10.10.10.5
PC5	10.10.10.6

5. Switch to Simulation Mode.
6. Send Simple PDU from 10.10.10.1 to 10.10.10.5.
7. Observe packet broadcasting.

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Part II – Switch

1. Add one Cisco 2960 Switch.
2. Add four PCs.
3. Connect using Straight-Through cables.
4. Observe MAC Address learning.
5. Verify communication using Ping.
6. Observe that packets are forwarded only to the destination port.

Part III – Router

1. Create LAN-1

Network:

10.0.0.0

Subnet Mask:

255.0.0.0

Assign

PC0 → 10.0.0.1

PC1 → 10.0.0.2

PC2 → 10.0.0.3

Router G0/0

10.0.0.4

2. Create LAN-2

Network

192.168.1.0

Subnet Mask

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255.255.255.0

Assign

PC3 → 192.168.1.1

PC4 → 192.168.1.2

PC5 → 192.168.1.3

Router G0/1

192.168.1.4

3. Configure Default Gateway

LAN-1

10.0.0.4

LAN-2

192.168.1.4

Verify communication using Ping.

PROCEDURE:

1. Hub

- Place Hub and six PCs.
- Connect using Straight-through cables.
- Assign IP addresses.
- Send Simple PDU.
- Observe broadcasting.

2. Switch

- Place one Switch.

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- Connect four PCs.
- Assign IP addresses.
- Verify MAC learning.
- Ping between PCs.

3. Router

- Create two LANs.
- Connect each LAN to Router.
- Configure router interfaces.
- Configure Default Gateway.
- Verify communication between LANs using Ping.
- 4. Layer-2 Switch

INPUT/OUTPUT DETAILS:

(a) Devices Used

- Hub-PT
- Cisco 2960 Switch
- Cisco 2911 Router
- PC-PT

(b) Cable Types

- Copper Straight-Through

(c) Router Interface Configuration

Interface	IP Address
GigabitEthernet0/0	10.0.0.4
GigabitEthernet0/1	192.168.1.4

(d) Network Details

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LAN-1

Network Address

10.0.0.0

Mask

255.0.0.0

Gateway

10.0.0.4

LAN-2

Network Address

192.168.1.0

Mask

255.255.255.0

Gateway

192.168.1.4

DEMONSTRATION/OUTPUT:

- The Hub broadcasts packets to all connected devices.
- The Switch forwards packets only to the required destination using its MAC Address Table.
- The Router successfully transfers packets between two different LANs using routing and default gateway configuration.
- Communication between devices was verified successfully using Ping and Simple PDU.

Part-1: Hub

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Hub

1. Hubs works at the physical layer of OSI model
2. Used to set up LAN
3. Hub has multiple ports
4. Star Toplogy is developed with hub

5. When a packet arrives at one port, it is copies to the other ports so that all the segments of the LAN can see all packets.

STEPS:

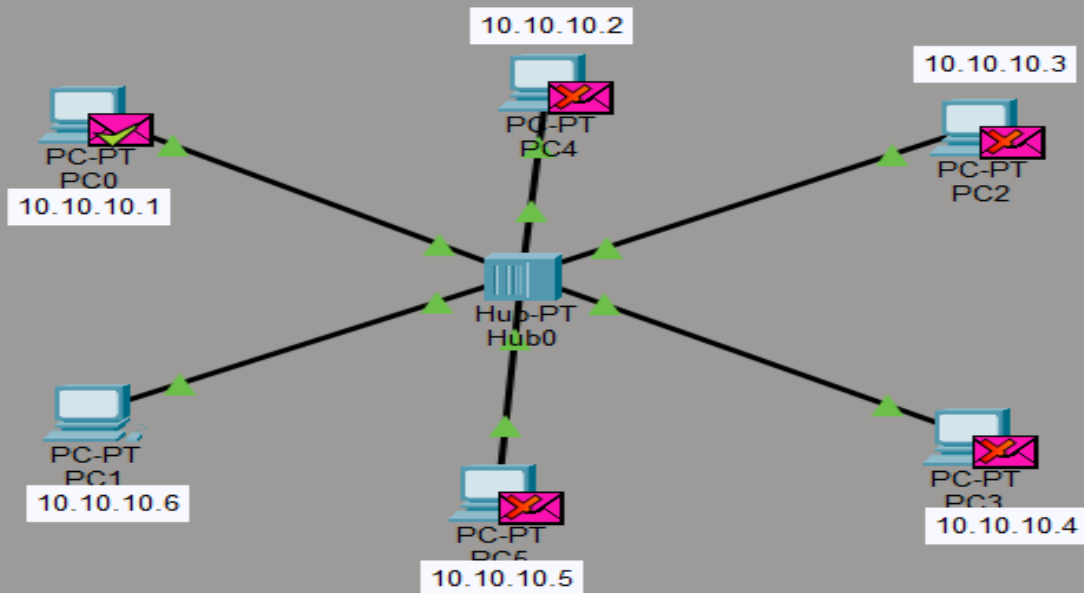
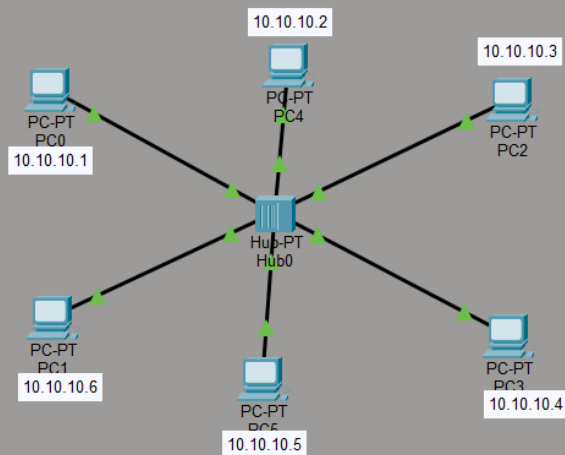
1. Hub section -> choose one hub -> click on hub and check for the ports
2. Take 6 PC's and take straight through cable -> Connect all PC's with Hub with FastEthernet port
3. On bottom right corner click on simulation -> take one data packet -> take 10.10.10.1 as Source_ip & 10.10.10.5 as Destination_ip

Advantage:

1. Cheaper than Switches.
2. Works good for smaller networks.

Disadvantages:

1. Issues with broadcasting.
2. No Memory.
3. Normally runs in half-duplex mode.



Part-2: Switch

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```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

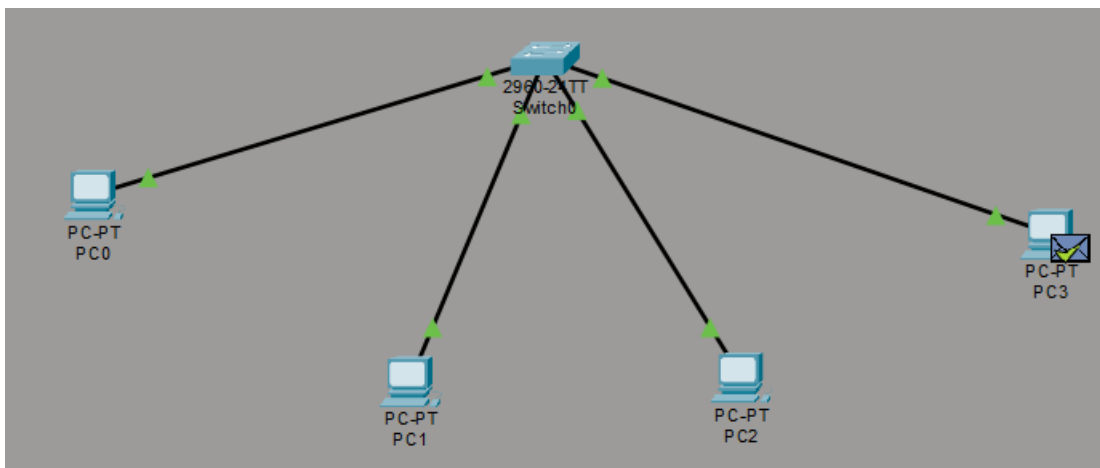
C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.3

Pinging 10.10.10.3 with 32 bytes of data:

Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time=1ms TTL=128

Ping statistics for 10.10.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



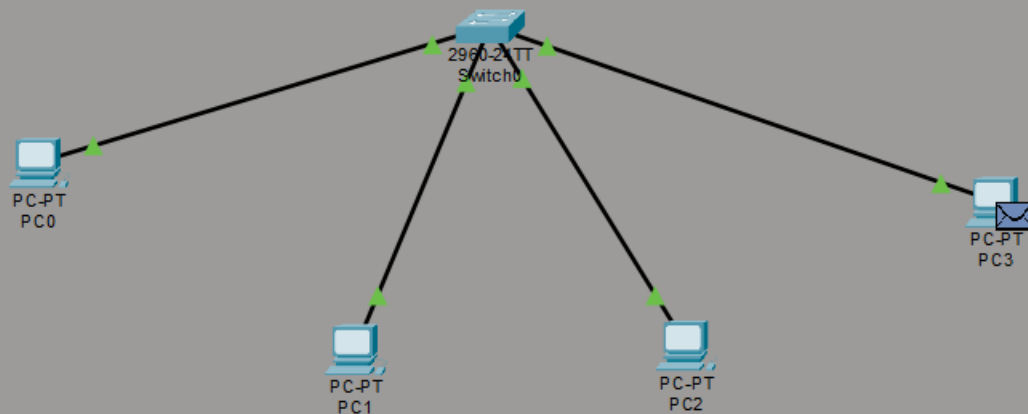
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Switches:

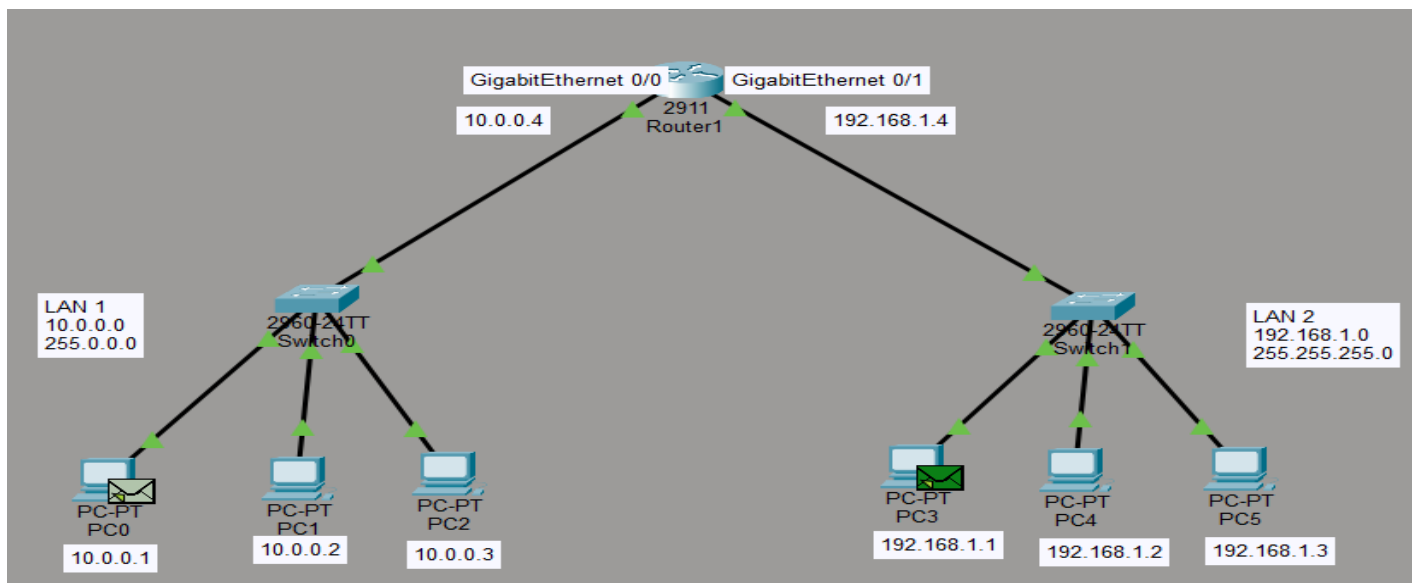
1. A switch is a networking device that connects devices on a computer network to establish a local area network.
2. Unlike hub, Switch has memory. [It stores the MAC Address table into its memory].
3. Store MAC Address Table.
4. It is layer 2 Device for setting up the LAN.
5. It has n no of ports which are called as Interface / Ports. [Every port is given a number so that we can identify who is connected to which port with MAC Address].

MAC Address Table looks like:

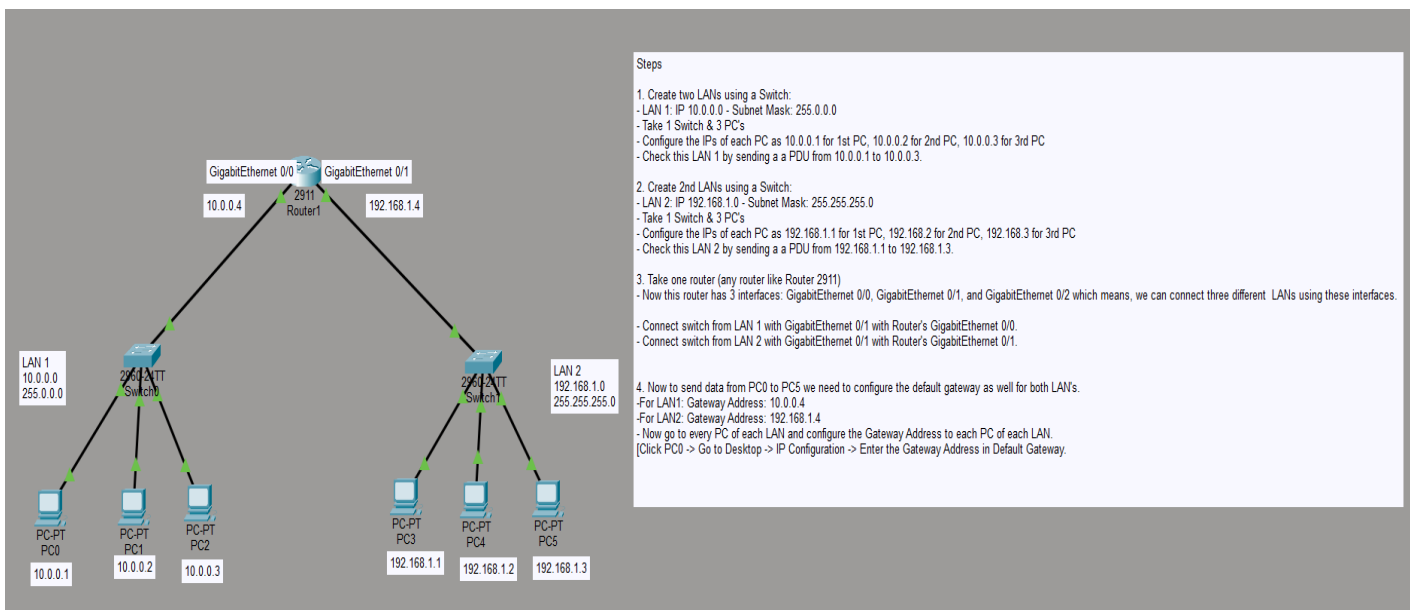
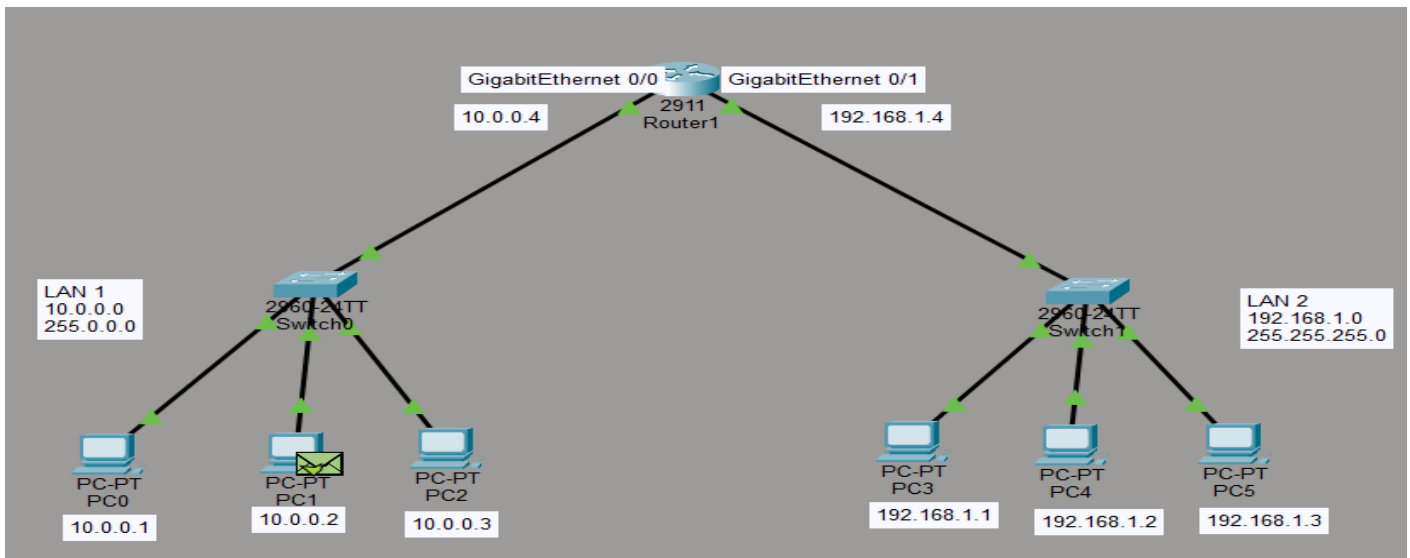
MAC Address	Interface / Port
AA-AA-AA-AA-AA-AA	1 [PC connected to port 1 of switch having MAC Address as AA-AA-AA-AA-AA-AA]
BB-BB-BB-BB-BB-BB	



Part-3: Router



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LEARNING OUTCOMES:

After completing this experiment:

- I understood the working principles of Hub, Switch, and Router by studying their functions, operating layers in the OSI model, and roles in establishing communication within and between computer networks.
- I learnt about the configuration and implementation of Local Area Networks (LANs) using Cisco Packet Tracer by connecting multiple PCs with hubs and switches, assigning valid IPv4

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addresses and subnet masks, and verifying successful communication using the Ping command and Simple PDU simulation.

3. I developed practical knowledge of data forwarding mechanisms by observing that a Hub broadcasts data to all connected devices, a Switch forwards frames using its MAC Address Table, and a Router forwards packets between different networks using IP addresses and a Routing Table.
4. I acquired hands-on experience in configuring routers by assigning IP addresses to router interfaces, configuring default gateways for different LANs, connecting multiple networks, and enabling communication between hosts located on separate subnets through proper routing.
5. I enhanced the ability to analyze, compare, and troubleshoot networking devices by understanding the differences between Hub, Switch, and Router in terms of OSI layer, addressing method, forwarding technique, memory, collision and broadcast domains, duplex mode, security, performance, and practical applications.